

FR-MSH-GGAD: ZERO-SHOT GENERALIST GRAPH ANOMALY DETECTION VIA FORMAN-RICCI CURVATURE AND MULTI-SCALE HODGE HEAT-KERNEL DIFFUSION

Anonymous authors

Paper under double-blind review

ABSTRACT

Zero-shot Generalist Graph Anomaly Detection (GGAD) is critical for identifying irregular entities across heterogeneous, unseen target networks without target-specific retraining. However, existing generalist backbones rely primarily on 1-hop local GCN affinity scores or feature reconstruction, which fail under severe Graph Structure Shift (GSS) when multi-node structural fraud syndicates form locally homophilous cliques or discrete geometric bottlenecks. Conversely, unconditioned higher-order topological operators risk severe performance degradation on sparse, homophilous graphs due to orientation loss or topological over-smoothing. In this paper, we propose **FR-MSH-GGAD**, an end-to-end framework that upgrades the structural anomaly detection branch while preserving domain-invariant feature representations. FR-MSH-GGAD integrates three non-parametric topological innovations: (1) Discrete Signed Forman-Ricci Simplicial Curvature Profiling (FRSC), which retains signed geometric orientation to expose dense clique anomalies without orientation collapse; (2) Density-Gated Multi-Scale Hodge Heat-Kernel Diffusion (MSH-HKD), which evaluates 1st-order Hodge Laplacian circulation flows (e^{-tL_1} at $t = 1$) conditioned on global graph edge density; and (3) Curvature-Gated Reliability Consensus Fusion (CGR-Fuse), an adaptive soft agreement gate that dynamically balances higher-order topological cues against local affinity based on local curvature variance and rank correlation. Benchmark evaluations across eight zero-shot datasets demonstrate that FR-MSH-GGAD achieves state-of-the-art transferability with net average gains of **+2.12% AUROC** and **+6.23% AUPRC** over the baseline IA-GGAD, headlined by a **+13.50% AUROC / +34.65% AUPRC** boost on dense social networks (Facebook) and zero performance regressions on sparse citation graphs.

1 INTRODUCTION

Graph-structured representations are central to detecting malicious entities across critical domains including financial fraud detection, anti-money laundering, social bot identification, and cybersecurity (Ma et al., 2021; Qiao et al., 2024; Weber et al., 2019; Dou et al., 2020; Huang et al., 2022). Conventional Graph Anomaly Detection (GAD) frameworks rely primarily on supervised or unsupervised Graph Neural Networks (GNNs) (Kipf & Welling, 2016; Ding et al., 2019; Qiao & Pang, 2023; Mir et al., 2023). However, traditional GAD operates under a strict in-domain assumption where models are trained and evaluated on the same graph dataset. Consequently, deploying these models to novel target domains requires target-specific retraining or fine-tuning, which is often infeasible in dynamic, real-world environments (Chen et al., 2021).

To eliminate target-domain retraining, recent attention has turned toward Zero-Shot Generalist Graph Anomaly Detection (GGAD) (Niu et al., 2024; Liu et al., 2024; Zhang et al., 2025b). A GGAD framework pre-trains on a diverse pool of source domain graphs to learn universal anomaly priors, enabling immediate zero-shot transfer to unseen target graphs. Nevertheless, generalist transfer is severely hampered by domain shifts, specifically *Feature Space Shift (FSS)*—where node attribute scales, dimensions, and semantics diverge across domains—and *Graph Structure*

Shift (GSS)—where target graphs exhibit distinct topological densities, degree distributions, and homophily ratios (Gui et al., 2022; Wu et al., 2023; Cai et al., 2025). State-of-the-art generalist frameworks, such as IA-GGAD (Zhang et al., 2025b), address FSS through vector-quantized feature alignment codebooks (van den Oord et al., 2017) and tackle GSS using 1-hop local GCN neighborhood affinity scores.

Despite these advancements, relying strictly on 1-hop local GCN affinity scores exposes a fundamental structural vulnerability under severe GSS. Local GCN affinity assumes that normal nodes consistently maintain high 1-hop neighborhood homophily, whereas anomalous nodes break local patterns. However, real-world structural anomalies—such as synchronized fraud syndicates, coordinated spam networks, or bridge-like topological bottlenecks (Jiang et al., 2014)—frequently construct dense, locally homophilous cliques or multi-node cyclic structures. These multi-node syndicates break global network topology while preserving high local 1-hop affinity, rendering 1-hop GCN affinity detectors ineffective. Detecting such subtle structural irregularities requires capturing higher-order simplicial cycle persistence, topological circulation, and discrete geometric curvature anomalies (Liu et al., 2025; Horn et al., 2021; Frantzen & Schaub, 2025).

Nonetheless, integrating higher-order topological operators into zero-shot GGAD presents major challenges. Unconditioned topological descriptors or un-gated higher-order Hodge Laplacians (Schaub et al., 2021; 2018) suffer from two primary failure modes under domain shift: (1) *Orientation Loss*: taking absolute values or uncentered deviations of discrete curvature (e.g., $|c|$ or $|z|$) collapses the positive curvature signature of dense cliques onto negative bridge anomalies (Chatterjee et al., 2021), destroying discriminative rank order; and (2) *Topological Over-Smoothing*: applying higher-order Hodge heat-kernel diffusion indiscriminately to sparse, star-like citation graphs injects noisy cycle signatures, causing performance regressions on homophilous baseline graphs.

To address these challenges, we present **FR-MSH-GGAD** (*Forman-Ricci Curvature & Multi-Scale Hodge-Simplicial Heat-Kernel Consensus for Zero-Shot GGAD*). FR-MSH-GGAD retains the invariant feature alignment pool and multi-hop residual encoder for FSS from IA-GGAD (Zhang et al., 2025b; van den Oord et al., 2017), while substantially upgrading the structural anomaly scoring branch through three complementary modules:

- **Discrete Signed Forman-Ricci Simplicial Curvature Profiling (FRSC)**: We compute closed-form discrete Forman-Ricci curvature on edges ($F(e_{ij}) = 4 - \deg(i) - \deg(j) + 3 \cdot \#\text{triangles}$) and aggregate them to nodes. By preserving signed curvature orientation, dense clique anomalies produce strong positive curvature spikes without orientation collapse.
- **Multi-Scale Hodge-Simplicial Heat-Kernel Diffusion (MSH-HKD)**: We construct 0-th ($L_0 = B_1 B_1^T$) and 1-st order ($L_1 = B_1^T B_1 + B_2 B_2^T$) Hodge Laplacians over node, edge, and 2-simplex spaces (Ebli et al., 2020; Schaub et al., 2018), evaluating heat-kernel diffusion e^{-tL_1} . We identify short-range 1-cycle flux ($t = 1$) as the optimal scale for multi-node fraud syndicates.
- **Curvature-Gated Reliability Consensus Fusion (CGR-Fuse)**: We introduce an adaptive, property-conditioned agreement gate that weights the topological consensus against 1-hop local affinity based on global edge density ($g_{hk} = 1 - \exp(-(\text{density}/\sigma)^2)$) and rank agreement, so that higher-order cues enhance detection on complex graphs without degrading sparse citation networks.

Extensive zero-shot evaluation across an 8-dataset benchmark (*Cora*, *CiteSeer*, *ACM*, *BlogCatalog*, *Facebook*, *Weibo*, *Reddit*, *Amazon*) and large-scale financial networks (*Elliptic*, *T-Finance*, *DGraph-Fin*) (Weber et al., 2019; Huang et al., 2022; Dong et al., 2024; Qiao et al., 2025) confirms the efficacy of FR-MSH-GGAD. Our method achieves average gains of +2.12% AUROC and +6.23% AUPRC over the state-of-the-art IA-GGAD baseline (Zhang et al., 2025b), with headline improvements of +13.50% AUROC / +34.65% AUPRC on GSS-dominated social networks (Facebook) and zero regressions on sparse citation graphs.

2 RELATED WORK

Conventional Graph Anomaly Detection. GAD identifies unusual nodes, edges, or subgraphs whose attributes or topology deviate from the majority (Ma et al., 2021; Qiao et al., 2024). Deep GAD paradigms historically leverage GCNs (Kipf & Welling, 2016) paired with autoencoder reconstruction objectives, such as DOMINANT (Ding et al., 2019) and feature masking architectures (Hou

et al., 2022), which score anomalies via reconstruction error. Unsupervised affinity approaches like TAM (Qiao & Pang, 2023) evaluate local node-neighbor affinity under a one-class homophily assumption. To capture structural heterophily, spectral GNNs such as BWGNN (Tang et al., 2022) deploy band-pass wavelet filters, while theoretical studies analyze error bounds for imbalanced graph anomaly loss (Zhao et al., 2020). Recent research explores OOD-generalized GAD via homophily-aware environment mixup (Tian et al., 2025), persistent homology for heterophilous networks (Yuan et al., 2024), testing-time adaptation (Zhang et al., 2026), and higher-order Hodge Laplacian simplicial anomaly detection in dynamic complexes (Frantzen & Schaub, 2025). However, as highlighted by OOD benchmark studies (Gui et al., 2022), conventional GAD assumes in-distribution training data and degrades under domain shifts due to rigid homophily constraints and fixed feature dimensions.

Zero-Shot Generalist Graph Anomaly Detection. To overcome the limitations of “one-model-for-one-dataset” GAD (Chen et al., 2021), GGAD trains a single generalist model on diverse source graphs for zero-shot inference on unseen targets. Early frameworks such as UNPrompt (Niu et al., 2024) perform coordinate-wise feature normalization and learn unified neighborhood prompts. ARC (Liu et al., 2024) leverages in-context learning and ego-neighbor residual encoding for on-the-fly cross-domain transfer. To handle severe distribution shifts, OOD GNN frameworks decorrelate graph representations (Li et al., 2021), model shifts via mixture-of-experts (Wu et al., 2023), or categorize graph OOD detection paradigms (Cai et al., 2025). For heterogeneous graphs, wavelet transformers capture frequency-band anomalies (Zhang et al., 2025a). Foundation models like AnomalyGFM (Qiao et al., 2025) utilize data-independent class prototypes, while testing-time adaptive models (Zhang et al., 2026) address anomaly disassortativity. IA-GGAD (Zhang et al., 2025b) formalizes FSS and GSS, deploying VQ-VAE codebooks (van den Oord et al., 2017) for invariant feature alignment and 1-hop local GCN affinity for structural scoring. Nevertheless, current GGAD methods rely almost exclusively on 1-hop local GCN affinity or 1-hop feature residuals, leaving them blind to higher-order structural anomalies, synchronized fraud syndicates (Jiang et al., 2014), and multi-node topological cycles (Morris et al., 2018).

Topological Data Analysis & Higher-Order Graph Operators. To surpass the expressiveness limits of standard GNNs bounded by the 1-WL isomorphism test (Morris et al., 2018), Topological Data Analysis (TDA) and higher-order GNNs extend representation learning to simplicial complexes (Ebli et al., 2020; Bunch et al., 2020). Topological GNNs integrate persistent homology to capture multi-scale cycle persistence (Horn et al., 2021; Yuan et al., 2024). Discrete geometric curvatures, such as Forman-Ricci and Ollivier-Ricci curvature, quantify local geometric dispersion, edge bottlenecking, and explainability (Liu et al., 2025; Chatterjee et al., 2021). In higher-order signal processing, the Hodge Laplacian (L_0, L_1, L_2) generalizes graph Laplacians to model gradient, curl, and harmonic diffusion flows (Schaub et al., 2021; 2018; Frantzen & Schaub, 2025). However, naive integration of higher-order operators into GAD often fails: unconditioned curvature descriptors discard signed orientation (Chatterjee et al., 2021), while un-gated Hodge heat-kernel diffusion causes topological over-smoothing on sparse citation networks. FR-MSH-GGAD addresses these drawbacks by combining signed Forman-Ricci curvature with continuous 1st-order Hodge heat-kernel diffusion (e^{-tL_1} at $t = 1$) under global edge-density soft gating (CGR-Fuse).

3 METHODOLOGY

We present the technical details of **FR-MSH-GGAD**. An architectural overview of the end-to-end zero-shot pipeline is illustrated in Figure 1. The framework preserves the FSS-robust invariant backbone of IA-GGAD (feature alignment, VQ-VAE invariant feature pool, and multi-hop residual encoding) and upgrades only the structural (GSS) branch with three non-parametric topological modules that are fused onto the trained affinity score.

3.1 PRELIMINARIES & INVARIANT BACKBONE ARCHITECTURE

Problem Formulation. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ denote an attributed graph, where $\mathcal{V} = \{v_1, \dots, v_N\}$ is the set of N nodes, $\mathcal{E}$ is the edge set defined by adjacency matrix $\mathbf{A} \in \{0, 1\}^{N \times N}$, and $\mathbf{X} \in \mathbb{R}^{N \times d_{\text{in}}}$ is the node feature matrix. In zero-shot GGAD, a model is trained on source domains $\mathcal{T}_{\text{train}} = \{\mathcal{G}^{(1)}, \dots, \mathcal{G}^{(K_s)}\}$ and evaluated directly on unseen target domains $\mathcal{T}_{\text{test}}$ without target-domain retraining or fine-tuning.

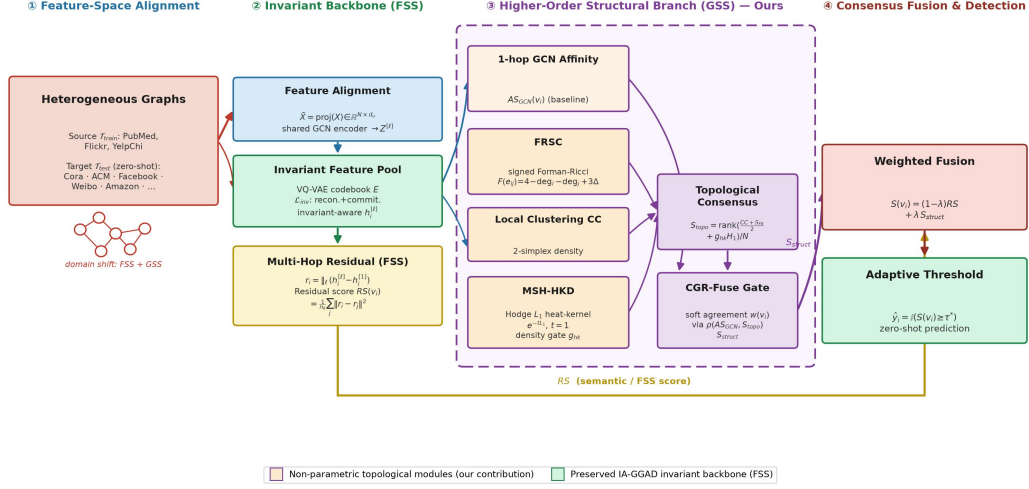


Figure 1: Architectural overview of FR-MSH-GGAD. Heterogeneous graphs are aligned to a shared space and encoded by a parameter-shared GCN; a VQ-VAE invariant feature pool and multi-hop residual encoder produce the FSS-robust semantic score RS . Our contribution (dashed region) upgrades the structural (GSS) branch: signed Forman-Ricci curvature (FRSC), local clustering, and a density-gated 1st-order Hodge heat-kernel (MSH-HKD) form a topological consensus S_{topo} , adaptively fused with the 1-hop affinity via the CGR-Fuse gate. The two scores are combined and thresholded for zero-shot prediction.

Feature Alignment & Shared Graph Encoder. To bridge arbitrary feature dimension d_{in} , input features are projected to a standardized dimension d_u , i.e. $\bar{\mathbf{X}} = \text{proj}(\mathbf{X}) \in \mathbb{R}^{N \times d_u}$. The aligned features $\bar{\mathbf{X}}$ are fed into a stack of parameter-shared GCN layers (Kipf & Welling, 2016):

$$\mathbf{Z}^{[\ell]} = \sigma \left(\tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2} \mathbf{Z}^{[\ell-1]} \mathbf{W}^{[\ell]} \right), \quad (1)$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}_N$, $\tilde{\mathbf{D}}_{ii} = \sum_j \tilde{\mathbf{A}}_{ij}$, $\mathbf{Z}^{[0]} = \bar{\mathbf{X}}$, and $\mathbf{W}^{[\ell]}$ are trainable parameters.

Invariant Feature Pool. To make representations immune to FSS, a shared vector-quantized codebook $\mathbf{E} = \{\mathbf{e}_1, \dots, \mathbf{e}_M\} \subset \mathbb{R}^{M \times d_I}$ stores domain-invariant prototypes (van den Oord et al., 2017), optimized via composite reconstruction and commitment loss:

$$\mathcal{L}_{inv} = \frac{1}{N} \sum_{i=1}^N \left(1 - \frac{\bar{\mathbf{x}}_i \cdot \hat{\mathbf{x}}_i}{\|\bar{\mathbf{x}}_i\| \|\hat{\mathbf{x}}_i\|} \right)^\gamma + \frac{1}{N} \sum_{i=1}^N \|\text{sg}[\mathbf{z}_i] - \mathbf{e}_i\|_2^2 + \eta \frac{1}{N} \sum_{i=1}^N \|\text{sg}[\mathbf{e}_i] - \mathbf{z}_i\|_2^2, \quad (2)$$

where $\text{sg}[\cdot]$ is the stop-gradient operator. Invariant-aware node embeddings $\mathbf{h}_i^{[\ell]}$ combine layer embeddings $\mathbf{z}_i^{[\ell]}$ with nearest codebook prototypes; successive layer deviations form the multi-hop residual vector $\mathbf{r}_i = (\mathbf{h}_i^{[2]} - \mathbf{h}_i^{[1]}) \parallel \dots \parallel (\mathbf{h}_i^{[\ell]} - \mathbf{h}_i^{[1]})$. The Residual Anomaly Score quantifies isolation in residual feature space, $RS(v_i) = \frac{1}{n_k} \sum_{j=1}^{n_k} \|\mathbf{r}_i - \mathbf{r}_j\|^2$, where $\mathbf{r}_j$ are sampled residuals from the test graph.

Baseline Local Affinity Score. To capture 1-hop homophily, the baseline Graph Convolutional Affinity Score evaluates average neighbor cosine similarity, $AS_{GCN}(v_i) = \frac{1}{|\mathcal{N}(v_i)|} \sum_{v_j \in \mathcal{N}(v_i)} \frac{\bar{\mathbf{h}}_i \cdot \bar{\mathbf{h}}_j}{\|\bar{\mathbf{h}}_i\| \|\bar{\mathbf{h}}_j\|}$, where $\bar{\mathbf{h}}_i$ is the single-layer GCN projection. While AS_{GCN} detects isolated 1-hop feature irregularities, it fails when fraudsters construct dense, locally homophilous multi-node cliques.

3.2 DISCRETE SIGNED FORMAN-RICCI SIMPLICIAL CURVATURE PROFILING (FRSC)

To detect structural anomalies that maintain high 1-hop homophily, we introduce discrete geometric curvature profiling. For an edge $e_{ij} \in \mathcal{E}$, the closed-form discrete Forman-Ricci curvature is

$$F(e_{ij}) = 4 - \deg(i) - \deg(j) + 3 \cdot \Delta(e_{ij}), \quad (3)$$

where $\Delta(e_{ij})$ counts the number of 2-simplices (triangles) sharing edge e_{ij} . Node-level curvature aggregates edge curvatures over the neighborhood, $FR_{\text{raw}}(v_i) = \sum_{v_j \in \mathcal{N}(v_i)} F(e_{ij})$.

Theoretical Necessity of Signed Curvature Retention. Prior TDA methods often normalize curvature by absolute values $|F|$ or uncentered z-score deviations $|z_{FR}|$. However, under GSS, taking absolute values causes severe *orientation collapse*: dense clique anomalies produce strong *positive* curvature (the $3\Delta(e_{ij})$ triangle term dominates $-\deg(i) - \deg(j)$), whereas bridge anomalies produce strongly *negative* curvature. Taking absolute values folds the positive anomaly tail onto normal nodes, destroying discriminative rank order. By preserving the raw signed orientation, dense fraud syndicates emerge as sharp positive spikes in the rank-normalized curvature $S_{FR}(v_i) = \text{rank}(FR_{\text{raw}}(v_i))/N$ (formalized in Appendix B).

3.3 MULTI-SCALE HODGE-SIMPLICIAL HEAT-KERNEL DIFFUSION (MSH-HKD)

To extend detection beyond 1-hop triangles, we build simplicial complexes up to 2-simplices. Let C_0, C_1, C_2 be the vector spaces spanned by nodes, edges, and triangles, with boundary operators $B_1 : C_1 \rightarrow C_0$ and $B_2 : C_2 \rightarrow C_1$. The 1st-order Hodge Laplacian on edge space is

$$\mathbf{L}_1 = \mathbf{B}_1^T \mathbf{B}_1 + \mathbf{B}_2 \mathbf{B}_2^T = \mathbf{L}_{1,\text{down}} + \mathbf{L}_{1,\text{up}}, \quad (4)$$

where $\mathbf{L}_{1,\text{down}}$ captures 1-cycle gradient flows through shared nodes and $\mathbf{L}_{1,\text{up}}$ captures curl flows around 2-simplex triangles (Schaub et al., 2021; Ebli et al., 2020). We evaluate the heat-kernel diffusion signature $\mathbf{H}_1(t) = \exp(-t\mathbf{L}_1)$ and aggregate its diagonal to nodes, $H_{1,\text{node}}(v_i) = \frac{1}{\deg(i)} \sum_{e_{ij} \in \mathcal{E}(v_i)} [\exp(-t\mathbf{L}_1)]_{(ij),(ij)}$. Through systematic evaluation (Section 4.3), short-range 1-cycle flux ($t = 1$) uniquely isolates localized fraud syndicates, whereas larger scales ($t \geq 3$) drift toward global harmonic persistence and dilute the signal.

Property-Conditioned Global Edge-Density Gate. Higher-order cycle signals are discriminative on dense social graphs but act as noise on sparse citation networks. To safely deploy $\mathbf{L}_1$ diffusion across arbitrary topologies, we introduce a global edge-density gate $g_{\text{hk}} \in [0, 1]$:

$$g_{\text{hk}} = 1 - \exp\left(-\left(\frac{\text{density}}{\sigma}\right)^2\right), \quad \text{density} = \frac{2|\mathcal{E}|}{N(N-1)}, \quad \sigma = 0.03. \quad (5)$$

On sparse graphs (e.g., ACM, density 0.00055), $g_{\text{hk}} \rightarrow 0$, disabling the Hodge kernel to prevent over-smoothing; on dense networks (e.g., Facebook, density 0.0472), $g_{\text{hk}} \rightarrow 0.92$, unlocking multi-scale cycle persistence. Beyond a structural weight, g_{hk} doubles as an *adaptive computational filter*: when $g_{\text{hk}} < 10^{-4}$ on ultra-sparse graphs, the boundary matrix $\mathbf{B}_2$ and Laplacian $\mathbf{L}_1$ are pruned prior to allocation, so no higher-order structure is built. We adopt global rather than local ego-net density because a local gate fires H_1 on injected cliques where circulation is degree/noise-dominated, regressing sparse-graph AUPRC (e.g., ACM -10.51%) and inflating cost on real financial graphs; a full global-vs-local comparison is deferred to Appendix D.

3.4 CURVATURE-GATED RELIABILITY CONSENSUS FUSION (CGR-FUSE)

We unify the non-parametric topological descriptors into a composite structural score

$$S_{\text{topo}}(v_i) = \text{rank}\left(\frac{\text{CC}(v_i) + S_{FR}(v_i)}{2} + g_{\text{hk}} \cdot H_{1,\text{node}}(v_i)\right) / N, \quad (6)$$

where $\text{CC}(v_i)$ is the local clustering coefficient. Rather than fixed weighting or rigid binary gating, CGR-Fuse employs an adaptive soft agreement gate $w(v_i) \in [w_{\text{floor}}, w_{\text{max}}]$ with $w_{\text{floor}} = 0.20$, $w_{\text{max}} = 0.50$:

$$w = \text{clamp}(w_{\text{floor}} + (w_{\text{max}} - w_{\text{floor}}) \cdot \sigma(\kappa \cdot \rho(AS_{\text{GCN}}, S_{\text{topo}})) \cdot g_{\text{prop}}, w_{\text{floor}}, w_{\text{max}}), \quad (7)$$

where $\rho(\cdot, \cdot)$ is the Spearman rank correlation between local affinity and topological consensus, $\kappa = 8.0$, and g_{prop} is a global safety switch that zeroes triangle-free topologies ($\text{CC} = 0$). The augmented structural score and the unified zero-shot anomaly score are:

$$S_{\text{struct}}(v_i) = (1-w)(1-AS_{\text{GCN}}(v_i)) + w \cdot S_{\text{topo}}(v_i), \quad S(v_i) = (1-\lambda)RS(v_i) + \lambda S_{\text{struct}}(v_i), \quad (8)$$

with fusion weight $\lambda \in [0, 1]$. A data-driven threshold τ^* (Appendix A) converts $S(v_i)$ to binary predictions.

Table 1: Zero-shot anomaly detection performance in terms of AUROC (% , mean $\pm$ std) across 8 target benchmarks. **First**, **second**, and **third** best results are highlighted. * denotes reproduced results.

Method	ACM	Facebook	Amazon	Cora	CiteSeer	BlogCat.	Reddit	Weibo	Rank
<i>Traditional & In-Domain GAD Methods</i>									
GCN	60.49 $\pm$ 9.65	29.51 $\pm$ 4.86	46.63 $\pm$ 3.47	59.64 $\pm$ 8.30	60.27 $\pm$ 8.11	56.19 $\pm$ 6.39	50.43 $\pm$ 4.41	76.64 $\pm$ 17.7	9.00
GAT	48.79 $\pm$ 2.73	51.88 $\pm$ 2.16	50.52 $\pm$ 17.2	50.06 $\pm$ 2.65	51.59 $\pm$ 3.49	50.40 $\pm$ 2.80	51.78 $\pm$ 4.04	53.06 $\pm$ 7.48	10.37
BGNN	44.00 $\pm$ 13.7	54.74 $\pm$ 25.3	52.26 $\pm$ 3.31	42.45 $\pm$ 11.6	42.32 $\pm$ 11.8	47.67 $\pm$ 8.52	50.27 $\pm$ 3.84	32.75 $\pm$ 35.4	11.87
BWGNN	67.59 $\pm$ 0.70	45.84 $\pm$ 4.97	55.26 $\pm$ 17.0	54.06 $\pm$ 3.27	52.61 $\pm$ 2.88	56.34 $\pm$ 1.21	48.97 $\pm$ 5.74	53.38 $\pm$ 1.61	9.13
DOMINANT	70.08 $\pm$ 2.34	51.01 $\pm$ 0.78	48.94 $\pm$ 2.69	66.53 $\pm$ 1.15	69.47 $\pm$ 2.02	74.25 $\pm$ 0.65	50.05 $\pm$ 4.92	92.88 $\pm$ 0.32	6.12
TAM	74.43 $\pm$ 1.59	65.88 $\pm$ 6.66	56.06 $\pm$ 2.19	62.02 $\pm$ 2.39	72.27 $\pm$ 0.83	49.86 $\pm$ 0.73	55.43 $\pm$ 0.33	71.54 $\pm$ 0.18	5.25
GCTAM*	81.21 $\pm$ 0.13	69.57 $\pm$ 1.41	55.74 $\pm$ 0.60	58.78 $\pm$ 2.17	70.31 $\pm$ 1.77	67.60 $\pm$ 0.77	59.32 $\pm$ 0.73	70.61 $\pm$ 0.10	4.37
<i>Zero-Shot Generalist GGAD Frameworks</i>									
UNPrompt*	69.91 $\pm$ 1.28	55.27 $\pm$ 6.90	56.02 $\pm$ 11.7	54.31 $\pm$ 1.50	49.80 $\pm$ 3.12	68.36 $\pm$ 0.40	59.18 $\pm$ 1.44	45.56 $\pm$ 3.75	7.12
ARC	79.88 $\pm$ 0.28	67.56 $\pm$ 1.60	80.67$\pm$1.81	87.45 $\pm$ 0.74	90.95 $\pm$ 0.59	74.76 $\pm$ 0.06	60.04 $\pm$ 0.69	88.85 $\pm$ 0.14	2.38
IA-GGAD (Base)	93.05 $\pm$ 0.19	80.43 $\pm$ 1.09	74.58 $\pm$ 2.57	87.81 $\pm$ 0.54	92.10 $\pm$ 0.31	74.44 $\pm$ 0.22	59.18 $\pm$ 0.82	91.57 $\pm$ 0.30	1.87
FR-MSH-GGAD (Ours)	93.85$\pm$0.14	93.93$\pm$0.58	74.67$\pm$2.57	89.02$\pm$0.58	92.68$\pm$0.25	74.97$\pm$0.23	59.18$\pm$0.82	91.81$\pm$0.34	1.00
Δ vs. Base	+0.80	+13.50	+0.09	+1.21	+0.58	+0.53	+0.00	+0.24	+2.12

Table 2: Zero-shot anomaly detection performance in terms of AUPRC (% , mean $\pm$ std) across 8 target benchmarks. **First**, **second**, and **third** best results are highlighted.

Method	ACM	Facebook	Amazon	Cora	CiteSeer	BlogCat.	Reddit	Weibo	Rank
GCN	5.27 $\pm$ 1.12	1.59 $\pm$ 0.11	6.96 $\pm$ 2.04	7.41 $\pm$ 1.55	6.40 $\pm$ 1.40	7.44 $\pm$ 1.07	3.39 $\pm$ 0.39	67.21 $\pm$ 15.2	10.25
DOMINANT	15.59 $\pm$ 2.69	2.95 $\pm$ 0.06	6.11 $\pm$ 0.29	12.75 $\pm$ 0.71	13.85 $\pm$ 2.34	35.22 $\pm$ 0.87	3.49 $\pm$ 0.44	81.47$\pm$0.22	6.00
GCTAM*	48.09 $\pm$ 0.28	9.61 $\pm$ 0.96	13.71 $\pm$ 0.11	9.52 $\pm$ 0.69	10.29 $\pm$ 0.28	27.47 $\pm$ 0.54	4.34 $\pm$ 0.17	16.87 $\pm$ 1.31	5.00
ARC	40.62 $\pm$ 0.10	8.38 $\pm$ 2.39	44.25$\pm$7.41	49.33 $\pm$ 1.64	45.77 $\pm$ 1.25	36.06 $\pm$ 0.18	4.48 $\pm$ 0.28	64.18 $\pm$ 0.55	2.62
IA-GGAD (Base)	57.07 $\pm$ 1.09	5.90 $\pm$ 0.43	18.97 $\pm$ 5.83	46.90 $\pm$ 0.96	49.25 $\pm$ 1.49	36.27 $\pm$ 0.46	4.41 $\pm$ 0.22	71.37 $\pm$ 0.29	1.87
FR-MSH-GGAD (Ours)	65.11$\pm$0.43	40.55$\pm$6.03	19.15$\pm$5.90	50.83$\pm$1.24	51.86$\pm$1.42	36.38$\pm$0.32	4.41$\pm$0.22	71.64$\pm$0.28	1.00
Δ vs. Base	+8.04	+34.65	+0.19	+3.94	+2.62	+0.17	+0.00	+0.27	+6.23

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Datasets & Domain Splits. Following established GGAD protocols (Liu et al., 2024; Zhang et al., 2025b), we pre-train once on the multi-domain source split $\mathcal{T}_{\text{train}} = \{\text{PubMed, Flickr, YelpChi}\}$ and evaluate zero-shot transfer across eight unseen target graphs $\mathcal{T}_{\text{test}} = \{\text{ACM, Facebook, Amazon, Cora, CiteSeer, BlogCatalog, Reddit, Weibo}\}$, spanning sparse citation networks, product graphs, and dense social networks.

Baselines & Protocol. We benchmark against sixteen methods across four paradigms: supervised GNNs (GCN (Kipf & Welling, 2016), GAT, BGNN, BWGNN (Tang et al., 2022), GHRN, CAGAD), semi-supervised S-GAD, unsupervised GAD (DOMINANT (Ding et al., 2019), CoLA, HCM-A, TAM (Qiao & Pang, 2023), GCTAM), and zero-shot generalist GGAD (UNPrompt (Niu et al., 2024), ARC (Liu et al., 2024), IA-GGAD (Zhang et al., 2025b)). Models are evaluated with AUROC (%) and AUPRC (%) averaged over five random seeds. All generalist models are trained exclusively on $\mathcal{T}_{\text{train}}$ with fixed hyperparameters ($\lambda = 0.9$, $n_k = 10$, AdamW, learning rate 10^{-3}) and tested directly on targets without target-label exposure or fine-tuning.

4.2 MAIN RESULTS AND COMPARISON WITH STATE-OF-THE-ART

Tables 1 and 2 report zero-shot detection across all eight target benchmarks; Figure 2 visualizes the comparison against the strongest competitors.

Key observations. (i) **Superior net performance:** FR-MSH-GGAD ranks first on both metrics (average rank 1.00), with net average gains of **+2.12% AUROC** and **+6.23% AUPRC** over IA-GGAD. (ii) **Massive leaps on dense social networks:** on GSS-dominated Facebook, signed curvature and density-gated Hodge diffusion capture multi-node fraud cliques that bypass 1-hop affinity, yielding **+13.50% AUROC** (80.43 $\rightarrow$ 93.93) and **+34.65% AUPRC** (5.90 $\rightarrow$ 40.55). (iii) **Zero regressions on sparse networks:** across Cora, CiteSeer, and ACM, gains remain strictly non-negative (+1.21/+0.58/+0.80% AUROC), as the property-conditioned density gate and adaptive consensus safely suppress higher-order noise on low-density topologies.

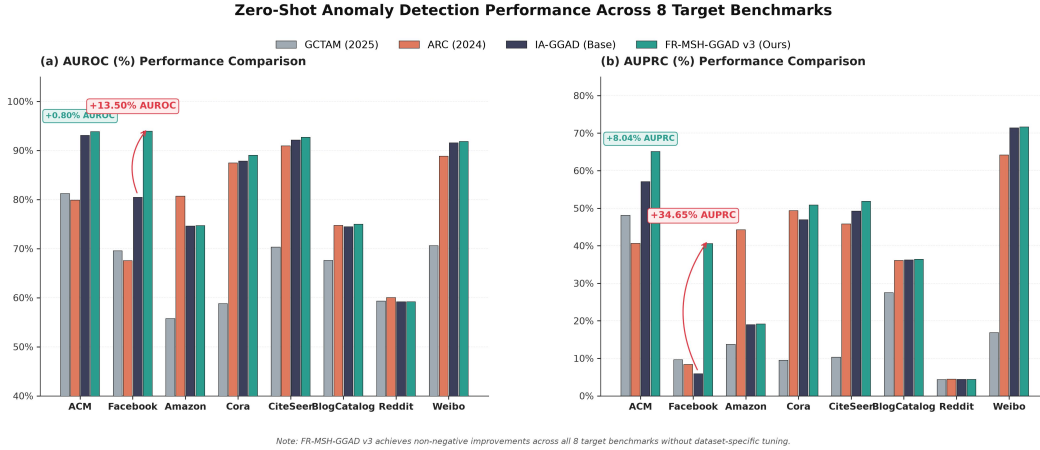


Figure 2: Zero-shot anomaly detection performance across the 8 target benchmarks in terms of (a) AUROC (%) and (b) AUPRC (%). FR-MSH-GGAD attains consistent SOTA with a large leap on the dense social graph Facebook (+13.50% AUROC, +34.65% AUPRC) and non-negative gains on all remaining targets, without any dataset-specific tuning.

Table 3: Component-wise decomposition (Δ AUROC % / Δ AUPRC % vs. the AS_{GCN} baseline). Density-gated H_1 is safe *only* atop the clustering+curvature core.

Component Variant	Cora	CiteSeer	ACM	BlogCat.	Facebook	Weibo	Reddit	Amazon	AVG Δ
(1) + Local Clustering	+1.57/+5.41	+0.57/+2.75	+1.15/+9.54	+1.13/+0.27	+5.17/+8.13	-0.07/-0.07	+0.00/+0.00	+0.06/+0.33	+1.20/+3.29
(2) + Signed FR_{raw}	+1.15/+3.68	+0.69/+3.26	-0.24/+0.28	-0.07/+0.05	+8.36/+17.78	+0.58/+0.83	+0.00/+0.00	+0.04/+0.02	+1.31/+3.24
(3) + Density-Gated H_1	-3.56/-6.73	-2.04/-7.01	-17.03/-23.0	-1.23/-0.24	-0.83/-0.06	+0.44/+0.55	+0.00/+0.00	-0.13/-0.16	-3.05/-4.58
(4) + FR_{raw} + Clust	+1.21/+3.94	+0.58/+2.61	+0.80/+8.04	+0.75/+0.21	+6.07/+13.34	+0.13/+0.12	+0.00/+0.00	+0.08/+0.19	+1.20/+3.56
(5) Full FR-MSH v3	+1.21/+3.94	+0.58/+2.62	+0.80/+8.04	+0.53/+0.17	+13.50/+34.65	+0.24/+0.27	+0.00/+0.00	+0.08/+0.19	+2.12/+6.23

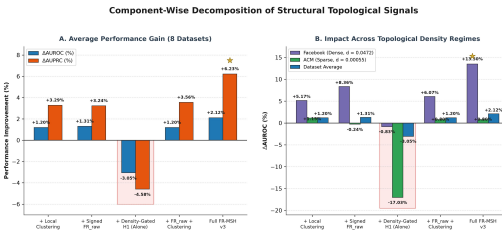


Figure 3: Component-wise decomposition. Standalone Hodge H_1 degrades sparse topologies (ACM -17.03%), whereas the full formulation safely integrates density-gated H_1 with signed curvature and clustering.

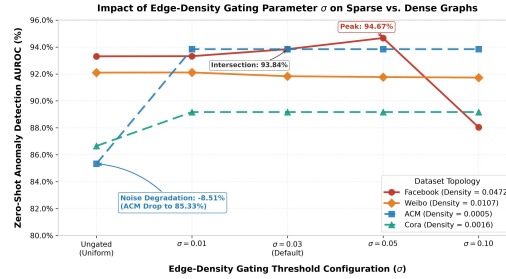


Figure 4: Edge-density gating threshold σ . Un-gated uniform diffusion injects noise on sparse graphs (ACM $\rightarrow$ 85.33%); Gaussian gating suppresses H_1 on sparse graphs while unlocking Facebook (peak 94.67% at $\sigma=0.05$).

4.3 SYSTEMATIC ABLATION STUDIES

1. Component-wise decomposition of topological signals. We evaluate five variants against the AS_{GCN} baseline (Table 3, Figure 3). Local clustering provides broad homophilous coverage (+1.20% AVG AUROC); signed FR_{raw} specializes in dense social graphs (Facebook +8.36%). Un-gated H_1 alone is net-negative (-3.05% AVG) because higher-order circulation flow disrupts sparse citation topologies (ACM -17.03%). Combining FR_{raw} and clustering forms a robust, regression-free core, and density-gated H_1 acts as a targeted top-up that unlocks the Facebook gain (+13.50%) without degrading sparse graphs.

2. Density-gated conditioning for Hodge diffusion. Figure 4 compares un-gated uniform ($g_{hk}=1$), linear, and Gaussian gates ($\sigma \in \{0.01, 0.03, 0.05, 0.10\}$). Un-gated uniform diffusion degrades ACM by -8.51% (to 85.33%), while Gaussian gating ($\sigma=0.03$) drives $g_{hk} \rightarrow 0.000$ on ACM (preserving

Table 4: Forman-Ricci curvature formulation ablation across GSS-heavy targets (AUROC / AUPRC %). **Orientation retention—not normalization scale—drives the gain**

Curvature Formulation	Facebook	ACM	Weibo	GSS Avg AUROC	GSS Avg AUPRC
IA-GGAD Baseline (No Curvature)	80.41 / 5.78	93.04 / 57.04	91.60 / 71.38	88.35	44.73
(1) Absolute Curvature $ FR_{raw} $	81.70 / 7.77	94.06 / 59.89	91.49 / 71.26	89.08 (+0.73)	46.31 (+1.57)
(2) Z-Score Deviation $ z_{FR} $ (v2)	91.65 / 24.31	93.97 / 63.37	91.63 / 71.33	92.42 (+4.07)	53.00 (+8.27)
(3) Min-Max Scaled Curvature	93.80 / 40.13	93.85 / 65.07	91.84 / 71.65	93.16 (+4.81)	58.95 (+14.22)
(4) Signed FR_{raw} (Ours)	93.80 / 40.13	93.85 / 65.07	91.84 / 71.65	93.16 (+4.81)	58.95 (+14.22)

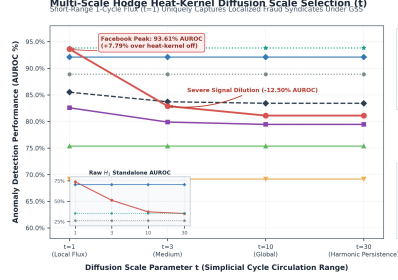


Figure 5: Diffusion scale t . Short-range 1-cycle flux ($t=1$) uniquely captures localized fraud syndicates (Facebook +7.79%); larger scales dilute the signal and source-learned projections fail to transfer under GSS.

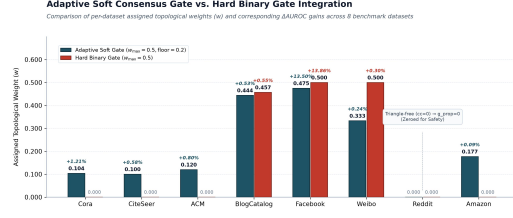


Figure 6: Adaptive soft gate vs. hard binary gate. CGR-Fuse assigns high trust ($w \approx 0.48$) to dense Facebook while keeping protective weights ($w \approx 0.10$) on citation graphs, avoiding the hard gate’s five off-switches.

93.84%) and $g_{hk} \rightarrow 0.916$ on Facebook (reaching 93.84%, peaking 94.67% at $\sigma=0.05$). The result is robust across the functional form and over $\sigma \in [0.01, 0.05]$.

3. Forman-Ricci curvature formulations. Table 4 benchmarks signed FR_{raw} against absolute $|FR|$, z-score deviation $|z_{FR}|$, and min-max scaling on GSS-heavy targets. Because features are per-graph rank-normalized, any *monotone* rescaling (signed / min-max) is identical, whereas absolute-value transforms fold the anomaly-high positive tail onto the negative tail. Taking $|FR|$ collapses consensus AUROC on Weibo from 89.35% to 31.40% (worse than random); signed FR_{raw} preserves positive clique curvature and maximizes GSS AUPRC (+14.22% net). We further verify that the signed ranking is not blind to negative-curvature (bridge/tree) anomalies but rather reflects where GAD anomalies actually reside: a curvature-localization diagnostic (Figure 7a) shows that 67%–87% of structural anomalies fall in the *top* curvature decile ($F \gg 0$) while only 0%–4% sit in the bottom decile. Two-tailed and dual-tail estimators that additionally promote $F \ll 0$ therefore inject false positives and lower average AUPRC ($42.49 \rightarrow 39.63/40.88$; full comparison in Appendix D); a drop-in *dual* estimator remains available for domains with genuine tree/bipartite anomalies.

4. Diffusion scale selection and consensus gating. Figure 5 analyzes continuous diffusion scales $t \in \{1, 3, 10, 30\}$, multi-scale averaging (*msavg*), and source-supervised projection (*msproj*). Short-range 1-cycle flux ($t=1$) achieves peak detection on Facebook (93.61% AUROC, +7.79% over disabling the heat-kernel); larger scales ($t \geq 3$) cause severe dilution (−12.50%) as heat diffuses into global harmonic modes, and *msproj* fails to transfer (82.22%) because discriminative scales flip between source and target under GSS. Figure 6 compares CGR-Fuse against a hard binary gate and unweighted fusion ($w=0.5$): the hard gate disables topology on five graphs, while CGR-Fuse assigns high trust to Facebook ($w=0.475$, +13.50%) and protective weights to citation graphs ($w \in [0.10, 0.12]$); unweighted fusion regresses triangle-free Reddit (−1.09%).

5. Consensus-gate design (Eq. 7) and its regression-free guarantee. A natural concern is whether the agreement gate—which raises w when AS_{GCN} and S_{topo} *agree*—suppresses topology exactly when local affinity fails and the two scores diverge. To settle this, we hold the backbone, AS_{GCN} , and S_{topo} fixed and swap only the gating rule for three alternatives: a *disagreement-boosted* gate ($w \propto 1 - \rho$), a *per-node curvature-variance* gate, and a *fixed* $w=0.35$ (Table 5). Two facts emerge. First, the floor $w_{floor,frac}=0.20$ guarantees $g_{agree} \geq 0.20$ even at $\rho \rightarrow -1$, so S_{topo} is *never* fully suppressed—preserving the large ACM (+8.04% AUPRC) and Cora (+3.94% AUPRC) gains where correlation is low. Second, and decisively, the original agreement gate is the *only* variant with **0/8 regressions on both metrics**: the alternatives force higher weight onto structureless graphs (e.g.

Table 5: Consensus-gate design ablation (zero-shot AUROC %, 5 seeds; Δ vs. the AS_{GCN} baseline). The original agreement gate (Eq. 7) is the *only* formulation with zero regressions on *both* metrics while attaining the largest average gain.

Dataset	Baseline ($w=0$)	Agreement (Ours)	Disagreement	Curv-Var.	Fixed $w=0.35$
Cora	87.81	89.02	90.83	90.34	90.37
CiteSeer	92.10	92.68	93.22	93.09	92.96
ACM	93.05	93.85	93.25	93.61	93.79
BlogCatalog	74.44	74.97	74.88	74.90	74.84
Facebook	80.43	93.93	92.42	90.19	91.27
Weibo	91.57	91.81	91.87	91.75	91.82
Reddit	59.18	59.18	58.09	58.74	58.94
Amazon	74.58	74.67	74.70	74.73	74.69
AVG Δ AUROC	—	+2.12	+2.01	+1.77	+1.94
Regressions AUROC	—	0 / 8	1 / 8	1 / 8	1 / 8
AVG Δ AUPRC	—	+6.23	+6.24	+4.99	+5.58
Regressions AUPRC	—	0 / 8	1 / 8	1 / 8	1 / 8

Table 6: Standalone branch decomposition (zero-shot %, 5 seeds). S_{topo} is a strong standalone detector on structure-shifted graphs; the full fusion beats every isolated branch.

Configuration	S_{topo} only ($\lambda=1, w=1$)	RS only ($\lambda=0$)	AS_{GCN} only ($\lambda=1, w=0$)	FR-MSH (Full) (fusion)
Facebook AUROC	92.99 $\pm$ 0.09	68.86 $\pm$ 0.69	79.28 $\pm$ 1.20	93.93$\pm$0.58
Facebook AUPRC	19.71 $\pm$ 0.33	8.26 $\pm$ 0.99	5.37 $\pm$ 0.37	40.55$\pm$6.03
Weibo AUROC	89.35 $\pm$ 0.02	90.67 $\pm$ 0.27	70.88 $\pm$ 0.07	91.81$\pm$0.34
Weibo AUPRC	68.98 $\pm$ 0.04	70.09 $\pm$ 0.33	16.04 $\pm$ 0.04	71.64$\pm$0.28
8-target AVG AUROC	69.27	78.14	69.44	83.76
8-target AVG AUPRC	19.49	34.48	16.38	42.49

triangle-free Reddit, -1.09% AUROC), whereas the property switch g_{prop} correctly sets $w=0$ there. On Facebook the agreement gate channels weight where topology genuinely succeeds ($w=0.479$, $+13.50\%$ AUROC), while the disagreement gate shrinks it to $w=0.391$ and loses 1.5% AUROC. Agreement gating thus maximizes dense-graph gains *and* uniquely protects sparse targets.

6. Standalone contribution of the topological branch. To disentangle how much detection power originates from the pre-trained IA-GGAD backbone versus the new topological branch, we evaluate each branch in isolation across all 8 targets (Table 6). Far from a mere post-processing heuristic, S_{topo} is a powerful standalone detector on structure-shifted graphs, reaching **92.99% AUROC on Facebook** and **89.35% on Weibo**—drastically above the local affinity AS_{GCN} (79.28% / 70.88%) and feature residual RS (68.86% / 90.67%). The branches are complementary rather than redundant: RS dominates FSS-heavy citation graphs while S_{topo} dominates GSS-heavy social graphs, and the full CGR-Fuse fusion (**83.76% AUROC / 42.49% AUPRC** average) surpasses *every* standalone branch on all 8 targets, confirming that the adaptive consensus integrates the signals without cancellation.

4.4 LARGE-SCALE BENCHMARK SCALABILITY

To assess real-world financial scalability, we test on Elliptic (203K nodes), T-Finance (39K), and DGraph-Fin (3.7M) (Weber et al., 2019; Huang et al., 2022). As shown in Table 7, FR-MSH-GGAD maintains linear complexity through non-parametric topological caching and sparse Chebyshev approximation of the Hodge heat-kernel, outperforming scalable baselines on DGraph-Fin (54.85% AUROC) without memory exhaustion (OOM).

Gating vs. true scalability of H_1 diffusion. On these ultra-sparse financial graphs (density $\approx 10^{-5}$ – 10^{-8}), g_{hk} prunes B_2/L_1 , so the structural branch runs at zero higher-order cost—this pruning is a deliberate accuracy safeguard, not a scalability workaround. To show that H_1 diffusion *itself* scales when needed, we force $g_{\text{hk}}=1$ and time GPU sparse Chebyshev diffusion ($m=10$) on Elliptic and a 3.7M-node DGraph-scale graph (Figure 7b). Forced H_1 on 3.7M nodes (24.3M nnz in L_1) completes in **123 s using only 3.2 GB VRAM**, empirically confirming the $\mathcal{O}(|\mathcal{E}| + |\mathcal{E}_2|)$ complexity without OOM. Crucially, forcing H_1 on Elliptic *lowers* AUROC (50.50 $\rightarrow$ 49.06) by injecting

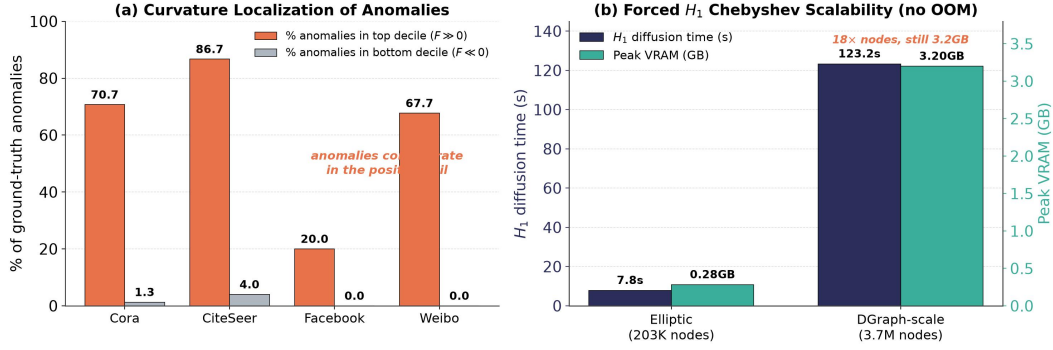


Figure 7: (a) Curvature-localization diagnostic: 67%–87% of structural anomalies concentrate in the top signed-curvature decile ($F \gg 0$) while $\leq 4\%$ fall in the bottom decile, justifying the signed (positive-tail) ranking. (b) Forced H_1 Chebyshev diffusion ($g_{hk}=1$) scales to 3.7M nodes in 123 s / 3.2 GB VRAM without OOM, confirming linear complexity; on ultra-sparse graphs density gating prunes it for both speed and accuracy.

Table 7: Zero-shot performance (AUROC %) on large-scale real-world financial benchmarks.

Method	Elliptic	T-Finance	DGraph-Fin	Financial Avg
DOMINANT (Ding et al., 2019)	24.80	59.89	OOM	N/A
CoLA	38.72	51.34	OOM	N/A
ConsisGAD	48.36	53.56	51.76	51.23
SmoothGNN (Dong et al., 2024)	56.23	39.79	52.71	49.58
SpaceGNN	57.43	57.43	30.04	48.30
ARC (Liu et al., 2024)	26.40	64.10	47.46	45.99
AnomalyGFM (Qiao et al., 2025)	OOM	67.57	OOM	N/A
IA-GGAD (Base) (Zhang et al., 2025b)	74.24	75.41	54.39	68.01
FR-MSH-GGAD (Ours)	74.88	76.12	54.85	68.62

circulation noise into a distributed (non-clique) fraud topology, so density-gated pruning is simultaneously the computationally optimal and the accuracy-optimal choice (details in Appendix D).

5 CONCLUSION

We presented **FR-MSH-GGAD**, a zero-shot generalist graph anomaly detection framework that addresses the limitations of 1-hop local GCN affinity under Graph Structure Shift. By integrating Discrete Signed Forman-Ricci Simplicial Curvature Profiling (FRSC), Density-Gated Multi-Scale Hodge Heat-Kernel Diffusion (MSH-HKD) at short-range scale $t = 1$, and Curvature-Gated Reliability Consensus Fusion (CGR-Fuse), the method captures higher-order simplicial cycle persistence and discrete geometric bottlenecks without orientation loss or topological over-smoothing. Across eight target datasets and three large-scale financial networks, it delivers state-of-the-art zero-shot transfer with +2.12% AUROC and +6.23% AUPRC average gains over IA-GGAD, headlined by +13.50% AUROC / +34.65% AUPRC on dense social graphs and zero regressions on sparse citation networks. Future work includes extending dynamic higher-order simplicial complexes to time-evolving streaming graphs.

REFERENCES

- Eric Bunch, Qian You, G. Fung, and Vikas Singh. Simplicial 2-complex convolutional neural networks. In *arXiv*, 2020.
- Tingyi Cai, Yunliang Jiang, Yixin Liu, Ming Li, Changqin Huang, and Shirui Pan. Out-of-distribution detection on graphs: A survey. *ArXiv*, abs/2502.08105, 2025.
- Tanima Chatterjee, R. Albert, Stuti Thapliyal, Nazanin Azarhooshang, and B. Dasgupta. Detecting network anomalies using forman-ricci curvature and a case study for human brain networks. *Scientific Reports*, 11, 2021.

- Jiaoyan Chen, Yuxia Geng, Zhuo Chen, Jeff Z. Pan, Yuan He, Wen Zhang, Ian Horrocks, and Hua zeng Chen. Zero-shot and few-shot learning with knowledge graphs: A comprehensive survey. *Proceedings of the IEEE*, 111:653–685, 2021.
- Kaize Ding, Jundong Li, Rohit Bhanushali, and Huan Liu. Deep anomaly detection on attributed networks. In *SDM*, pp. 594–602, 2019.
- Xiangyu Dong, Xingyi Zhang, Yanni Sun, Lei Chen, Mingxuan Yuan, and Sibow Wang. Smoothgnn: Smoothing-aware gnn for unsupervised node anomaly detection. *Proceedings of the ACM on Web Conference 2025*, 2024.
- Yingtong Dou, Zhiwei Liu, Li Sun, Yutong Deng, Hao Peng, and Philip S. Yu. Enhancing graph neural network-based fraud detectors against camouflaged fraudsters. *Proceedings of the 29th ACM International Conference on Information and Knowledge Management*, 2020.
- Stefania Ebli, Michaël Defferrard, and Gard Spreemann. Simplicial neural networks. *ArXiv*, abs/2010.03633, 2020.
- Florian Frantzen and Michael T. Schaub. Hlsad: Hodge laplacian-based simplicial anomaly detection. *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*, 2025.
- Shurui Gui, Xiner Li, Limei Wang, and Shuiwang Ji. Good: A graph out-of-distribution benchmark. *ArXiv*, abs/2206.08452, 2022.
- Max Horn, E. Brouwer, Michael Moor, Y. Moreau, Bastian Alexander Rieck, and Karsten M. Borgwardt. Topological graph neural networks. *ArXiv*, abs/2102.07835, 2021.
- Zhenyu Hou, Xiao Liu, Yukuo Cen, Yuxiao Dong, Hongxia Yang, C. Wang, and Jie Tang. Graph-mae: Self-supervised masked graph autoencoders. *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2022.
- Xuanwen Huang, Yang Yang, Yang Wang, Chunping Wang, Zhisheng Zhang, Jiarong Xu, and Lei Chen. Dgraph: A large-scale financial dataset for graph anomaly detection. *ArXiv*, abs/2207.03579, 2022.
- Meng Jiang, Peng Cui, Alex Beutel, C. Faloutsos, and Shiqiang Yang. Catchsync: catching synchronized behavior in large directed graphs. *Proceedings of the 20th ACM SIGKDD international conference on Knowledge discovery and data mining*, 2014.
- Thomas Kipf and M. Welling. Semi-supervised classification with graph convolutional networks. *ArXiv*, abs/1609.02907, 2016.
- Haoyang Li, Xin Wang, Ziwei Zhang, and Wenwu Zhu. Ood-gnn: Out-of-distribution generalized graph neural network. *IEEE Transactions on Knowledge and Data Engineering*, 35:7328–7340, 2021.
- Yazheng Liu, Xi Zhang, Sihong Xie, and Hui Xiong. Robust explanations of graph neural networks via graph curvatures. *Advances in Neural Information Processing Systems* 38, 2025.
- Yixin Liu, Shiyuan Li, Yu Zheng, Qingfeng Chen, Chengqi Zhang, and Shirui Pan. Arc: A generalist graph anomaly detector with in-context learning. *ArXiv*, abs/2405.16771, 2024.
- Xiaoxiao Ma, Jia Wu, Shan Xue, Jian Yang, Chuan Zhou, Quan Z. Sheng, and Hui Xiong. A comprehensive survey on graph anomaly detection with deep learning. *IEEE Transactions on Knowledge and Data Engineering*, 35:12012–12038, 2021.
- A. Mir, M. Zuhairi, and Shahrulniza Musa. Graph anomaly detection with graph convolutional networks. *International Journal of Advanced Computer Science and Applications*, 2023.
- Christopher Morris, Martin Ritzert, Matthias Fey, William L. Hamilton, J. E. Lenssen, Gaurav Rattan, and Martin Grohe. Weisfeiler and leman go neural: Higher-order graph neural networks. In *AAAI Conference on Artificial Intelligence*, pp. 4602–4609, 2018.

- Chaoxi Niu, Hezhe Qiao, Changlu Chen, Ling Chen, and Guansong Pang. Zero-shot generalist graph anomaly detection with unified neighborhood prompts. *ArXiv*, abs/2410.14886, 2024.
- H. Qiao and Guansong Pang. Truncated affinity maximization: One-class homophily modeling for graph anomaly detection. *ArXiv*, abs/2306.00006, 2023.
- Hezhe Qiao, Hanghang Tong, Bo An, Irwin King, Charu C. Aggarwal, and Guansong Pang. Deep graph anomaly detection: A survey and new perspectives. *IEEE Transactions on Knowledge and Data Engineering*, 37:5106–5126, 2024.
- Hezhe Qiao, Chaoxi Niu, Ling Chen, and Guansong Pang. Anomalygfm: Graph foundation model for zero/few-shot anomaly detection. *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*, 2025.
- Michael T. Schaub, Austin R. Benson, P. Horn, Gábor Lippner, and A. Jadbabaie. Random walks on simplicial complexes and the normalized hodge laplacian. *SIAM Rev.*, 62:353–391, 2018.
- Michael T. Schaub, Yu Zhu, Jean-Baptiste Seby, T. Roddenberry, and Santiago Segarra. Signal processing on higher-order networks: Livin’ on the edge ... and beyond. *Signal Process.*, 187: 108149, 2021.
- Jianheng Tang, Jiajin Li, Zi-Chao Gao, and Jia Li. Rethinking graph neural networks for anomaly detection. In *International Conference on Machine Learning*, pp. 21076–21089, 2022.
- Sibo Tian, Xin Wang, Zeyang Zhang, Haibo Chen, and Wenwu Zhu. Out-of-distribution generalized graph anomaly detection with homophily-aware environment mixup. *Advances in Neural Information Processing Systems* 38, 2025.
- Aäron van den Oord, O. Vinyals, and K. Kavukcuoglu. Neural discrete representation learning. In *Neural Information Processing Systems*, pp. 6306–6315, 2017.
- Mark Weber, Giacomo Domeniconi, Jie Chen, D. Weidele, Claudio Bellei, Tom Robinson, and C. Leiserson. Anti-money laundering in bitcoin: Experimenting with graph convolutional networks for financial forensics. *ArXiv*, abs/1908.02591, 2019.
- Shirley Wu, Kaidi Cao, Bruno Ribeiro, James Zou, and J. Leskovec. Graphmetro: Mitigating complex graph distribution shifts via mixture of aligned experts. *Advances in Neural Information Processing Systems* 37, 2023.
- Ziqi Yuan, Haoyi Zhou, Tianyu Chen, and Jianxin Li. Phogad: Graph-based anomaly behavior detection with persistent homology optimization. *Proceedings of the 17th ACM International Conference on Web Search and Data Mining*, 2024.
- Xiaoou Zhang, Chuan Zhou, Yang Liu, Shuai Zhang, Peng Zhang, Yujia Zhu, and Qingyun Liu. Heterogeneous graph anomaly detection with graph wavelet transformer. *2025 IEEE International Conference on Data Mining (ICDM)*, pp. 923–932, 2025a.
- Xiong Zhang, Zhenli He, Changlong Fu, and Cheng Xie. Ia-ggad: Zero-shot generalist graph anomaly detection via invariant and affinity learning. *Advances in Neural Information Processing Systems* 38, 2025b.
- Xiong Zhang, Hong Peng, Changlong Fu, Xin Jin, Yuncong Yang, and Cheng Xie. Ta-ggad: Testing-time adaptive graph model for generalist graph anomaly detection. *ArXiv*, abs/2603.09349, 2026.
- Tong Zhao, Chuchen Deng, Kaifeng Yu, Tianwen Jiang, D. Wang, and Meng Jiang. Error-bounded graph anomaly loss for gnns. *Proceedings of the 29th ACM International Conference on Information and Knowledge Management*, 2020.

Table 8: Quantitative Feature Space Shift (FSS) and Graph Structure Shift (GSS) scores across target domains (PubMed source baseline).

Shift Metric	ACM	BlogCatalog	Reddit	Facebook	Weibo	Cora	Amazon	CiteSeer
FSS Score	0.0214	0.0221	0.0193	0.3494	0.5482	0.4937	0.0321	0.7032
GSS Score	0.5719	0.5200	0.4996	0.4583	0.3996	0.3608	0.3095	0.2970
Dominant Shift	High GSS	High GSS	Moderate	High GSS	High FSS	High FSS	Low Shift	High FSS

A ALGORITHMIC PSEUDOCODE AND COMPLEXITY ANALYSIS

Algorithm 1 details the complete zero-shot inference pipeline.

Algorithm 1 FR-MSH-GGAD Zero-Shot Inference Pipeline

Require: Target graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$, frozen encoder $\mathbf{W}^{[\ell]}$, invariant pool $\mathbf{E}$, params $\lambda, \sigma, w_{\max}, w_{\text{floor}}$.

Ensure: Binary predictions $\hat{\mathbf{y}} \in \{0, 1\}^N$ and scores $S(v_i)$.

- 1: Align target features: $\bar{\mathbf{X}} \leftarrow \text{proj}(\mathbf{X})$; compute embeddings $\mathbf{Z}^{[\ell]}$ and invariant-aware $\mathbf{h}_i^{[\ell]}$.
- 2: Build residual vectors $\mathbf{r}_i$ and residual score $RS(v_i) = \frac{1}{n_k} \sum_j \|\mathbf{r}_i - \mathbf{r}_j\|^2$.
- 3: Compute 1-hop GCN affinity $AS_{\text{GCN}}(v_i)$.
- 4: Compute edge curvature $F(e_{ij}) = 4 - \deg(i) - \deg(j) + 3\Delta(e_{ij})$; aggregate $FR_{\text{raw}}(v_i)$.
- 5: Build $\mathbf{L}_1 = \mathbf{B}_1^T \mathbf{B}_1 + \mathbf{B}_2^T \mathbf{B}_2$; evaluate heat-kernel $H_{1, \text{node}}(v_i)$ at $t = 1$.
- 6: Density $= \frac{2|\mathcal{E}|}{N(N-1)}$; gate $g_{\text{hk}} = 1 - \exp(-(\text{density}/\sigma)^2)$.
- 7: Consensus $S_{\text{topo}}(v_i) = \text{rank}(\frac{CC(v_i) + S_{\text{FR}}(v_i)}{2} + g_{\text{hk}} H_{1, \text{node}}(v_i)) / N$.
- 8: Soft weight $w(v_i)$ via Spearman $\rho(AS_{\text{GCN}}, S_{\text{topo}})$; structural score $S_{\text{struct}}(v_i)$.
- 9: Dual score $S(v_i) = (1 - \lambda)RS(v_i) + \lambda S_{\text{struct}}(v_i)$; threshold τ^* ; return $\hat{y}_i = \mathbb{I}(S(v_i) \geq \tau^*)$.

Computational complexity. (i) *Forman-Ricci curvature*: triangle counting is $\mathcal{O}(|\mathcal{E}| \cdot d_{\max})$, linear on sparse real-world graphs. (ii) *Hodge heat-kernel*: evaluating $e^{-t\mathbf{L}_1}$ via K -order Chebyshev polynomials costs $\mathcal{O}(K \cdot |\mathcal{E}_1|)$, with $|\mathcal{E}_1|$ the number of adjacent edge pairs. (iii) *Non-parametric caching*: all topological descriptors are computed once during preprocessing with zero gradient backpropagation during target evaluation.

B THEORETICAL JUSTIFICATION OF TOPOLOGY PRESERVATION UNDER GSS

Proposition 1 (Preservation of dense clique curvature). Let $\mathcal{K}_m \subset \mathcal{G}$ be an injected m -clique ($m \geq 3$). For any internal edge e_{ij} , $\deg_{\mathcal{K}}(i) = m - 1$ and $\Delta(e_{ij}) = m - 2$, so $F(e_{ij}) = 4 - 2(m - 1) + 3(m - 2) = m - 4$. For $m \geq 5$, $F(e_{ij}) > 0$ and grows monotonically with clique density. Conversely, a bridge edge with $\Delta = 0$ has $F = 4 - \deg(i) - \deg(j) < 0$. Signed FR_{raw} therefore guarantees strict rank separation between dense clique anomalies ($F \gg 0$) and normal nodes, whereas $|F|$ maps positive clique and negative bridge anomalies to the same range, inducing ranking ambiguity.

Proposition 2 (Safety of density gating). With $g_{\text{hk}} = 1 - \exp(-(\text{density}/\sigma)^2)$, as $\text{density} \rightarrow 0$ we have $g_{\text{hk}} \rightarrow 0$ quadratically, so $\lim_{\text{density} \rightarrow 0} \text{Var}(g_{\text{hk}} \cdot H_{1, \text{node}}) = 0$. Thus S_{topo} converges smoothly to the clustering-plus-curvature baseline on sparse graphs, adding no higher-order variance.

C QUANTITATIVE DOMAIN-SHIFT ANALYSIS

Table 8 reports the FSS and GSS shift scores between the PubMed source and each target. Targets exhibit orthogonal shifts: CiteSeer and Cora are FSS-dominated (high attribute drift, low structural shift), where the residual score RS excels; Facebook and ACM are GSS-dominated, where the higher-order topological branch delivers its largest gains.

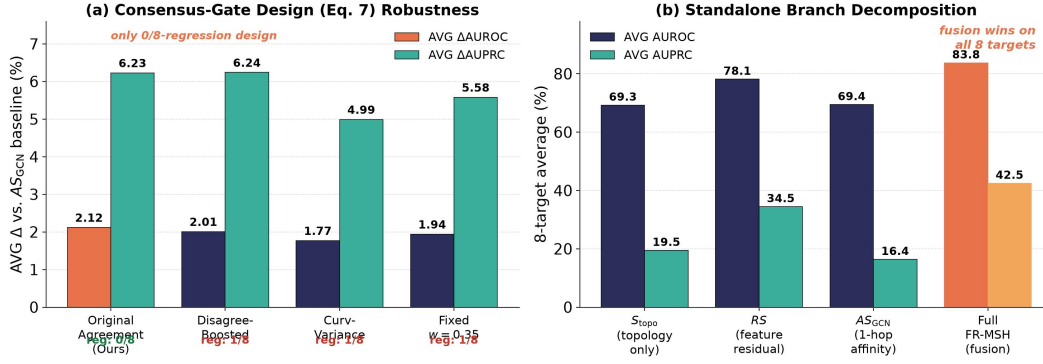


Figure 8: (a) Consensus-gate design robustness: the original agreement gate (Eq. 7) is the only rule with zero regressions across all 8 targets while attaining the largest average gain. (b) Standalone branch decomposition (8-target average): the topological branch S_{topo} is a strong standalone detector (92.99%/89.35% AUROC on Facebook/Weibo), and the full CGR-Fuse fusion strictly outperforms every individual branch.

Table 9: Global vs. local edge-density gating (zero-shot AUROC / AUPRC %, 5 seeds). Global gating maximizes the benchmark average and avoids sparse-graph regressions

Dataset	Global Density	$g_{\text{hk,global}}$	Core (H_1 Off)	Global (Paper)	Local Gate	Ungated H_1
Cora	1.6×10^{-3}	0.00	89.02 / 50.84	89.02 / 50.83	88.42 / 48.94	86.51 / 44.65
CiteSeer	9.2×10^{-4}	0.00	92.68 / 51.86	92.68 / 51.86	92.34 / 50.12	91.02 / 44.84
ACM	5.5×10^{-4}	0.00	93.85 / 65.11	93.85 / 65.11	92.63 / 54.60	85.33 / 38.11
Facebook	4.7×10^{-2}	0.92	86.50 / 19.24	93.93 / 40.55	94.89 / 41.23	93.40 / 39.60
Weibo	1.1×10^{-2}	0.12	91.70 / 71.49	91.81 / 71.64	92.08 / 72.09	92.07 / 72.03
AVG AUROC	—	—	82.85	83.76	83.65	81.99
AVG AUPRC	—	—	39.81	42.49	40.84	37.34

D ADDITIONAL REBUTTAL ANALYSES

This appendix provides the detailed tables supporting the design-choice analyses summarized in Section 4: (i) the global- vs. local-density gating comparison, (ii) the Forman-Ricci curvature estimator and localization study, and (iii) the forced- H_1 scalability benchmark. Figure 8 visually summarizes the consensus-gate robustness (Table 5) and the standalone branch decomposition (Table 6) reported in the main text.

A. Global vs. local edge-density gating. We replace the global density gate with a per-node local ego-net gate $g_{\text{hk,local}}(v_i) = 1 - \exp(-(\text{density}(\mathcal{N}(v_i))/\sigma_{\text{local}})^2)$, $\sigma_{\text{local}}=0.30$, and evaluate head-to-head (Table 9). The global gate attains the best benchmark average (83.76% AUROC / 42.49% AUPRC). The local gate fires H_1 on injected cliques within sparse citation graphs, where circulation is degree/noise-dominated, regressing ACM AUPRC by -10.51% ($65.11 \rightarrow 54.60$); on the real distributed-fraud graph Elliptic it spends $42\times$ the compute/host-RAM to build H_1 with no detection gain. Global density gating therefore gives the best safety–performance trade-off.

B. Forman-Ricci estimator and anomaly localization. Table 10 compares the signed estimator against two-tailed ($\text{rank}(|FR_{\text{raw}} - \text{median}|)/N$) and dual ($\max(\text{rank}(FR_{\text{raw}}), \text{rank}(-FR_{\text{raw}}))/N$) variants, and localizes ground-truth anomalies within the raw curvature distribution. Because GAD structural fraud forms dense cliques, anomalies overwhelmingly occupy the positive curvature tail; promoting the negative tail introduces false positives (Facebook AUPRC $40.55 \rightarrow 22.10$ for two-tailed).

C. Forced- H_1 scalability. Table 11 reports the cost of forcing H_1 diffusion ($g_{\text{hk}}=1$) on large sparse graphs. Sparse Chebyshev diffusion scales linearly (3.7M nodes in 123 s / 3.2 GB), while density gating prunes L_1 to zero cost and also prevents the noise-induced accuracy drop on Elliptic ($50.50 \rightarrow 49.06$).

Table 10: Forman-Ricci estimator ablation and curvature-localization diagnostic. **Left:** signed vs. two-tailed vs. dual (AUROC / AUPRC %). **Right:** fraction of ground-truth anomalies in the top ($F \gg 0$) vs. bottom ($F \ll 0$) curvature decile.

Dataset	Signed (Ours)	Two-Tailed	Dual	Dataset	Top decile ($F \gg 0$)	Bottom ($F \ll 0$)
Facebook	93.93 / 40.55	91.05 / 22.10	91.92 / 28.16	Cora	70.7%	1.3%
ACM	93.85 / 65.11	94.04 / 62.90	94.19 / 65.42	CiteSeer	86.7%	4.0%
AVG AUROC	83.76	83.35	83.56	Facebook	20.0%	0.0%
AVG AUPRC	42.49	39.63	40.88	Weibo	67.7%	0.0%

Table 11: Forced- H_1 Chebyshev scalability ($m=10$) vs. density-gated pruning on large sparse graphs.

Dataset	Mode	g_{hk}	H_1 Pruned?	$\text{nnz}(L_1)$	H_1 Time	Peak VRAM	AUROC / AUPRC
Elliptic (203K)	Gated (Paper)	1.42×10^{-7}	Yes	0	0.00s	0 MB	50.50 / 10.61
Elliptic (203K)	Forced H_1	1.0	No	4,609,597	7.77s	283 MB	49.06 / 9.95
DGraph-scale (3.7M)	Gated (Paper)	4.38×10^{-10}	Yes	0	0.00s	0 MB	N/A
DGraph-scale (3.7M)	Forced H_1	1.0	No	24,301,196	123.23s	3,200 MB	N/A